Course Code	**PRESTRESSED CONCRETE (PE – I)**	L	T	P	C
23A01504a		3	0	0	3

Course Objectives:-

The objectives of this course are to make the student to:

1. Understand the principles, methods, and materials used in prestressed concrete.
2. Analyze various losses of prestress in both pre-tensioned and post-tensioned members.
3. Design prestressed concrete beams considering flexure and shear forces.
4. Evaluate deflections in prestressed concrete structures and their controlling factors.
5. Analyze the behavior of composite beams under different loading conditions.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Explain the principles and methods of prestressing and the need for high-strength materials.
2. Analyze the different types of prestress losses and their impact on structural performance.
3. Design prestressed concrete beams considering flexural and shear stresses.
4. Evaluate deflections in prestressed beams and suggest control measures.
5. Analyze the stress distribution and differential shrinkage in composite beams.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO 1**	**PSO 2**
CO -1	2	1	1	-	-	1	-	-	-	1	-	1	1	1
CO -2	2	2	1	1	2	-	-	-	-	-	-	1	1	1
CO -3	2	2	3	2	2	-	1	-	-	-	-	1	2	2
CO -4	2	1	2	2	3	1	-	-	-	-	-	1	2	1
CO -5	2	2	2	1	2	1	-	-	-	-	-	1	2	1

UNIT – I

Introduction

Principles of Pre-Stressing – Prestressing Systems - Pre-Tensioning and Post Tensioning- Advantages and Limitations of Pre Stressed Concrete- Need for High Strength Materials. Methods of Pre-Stressing: Pre-Tensioning (Hoyer System) and Post-Tensioning Methods (Freyssinet System and Gifford- Udall System)

UNIT – II

Losses of pre-stress

Loss of Pre-Stress in Pre-Tensioned and Post-Tensioned Members Due to Elastic Shortening, Shrinkage and Creep of Concrete, Relaxation of Stress in Steel, Anchorage Slip and Frictional Losses.

UNIT – III

Flexural and Shear

Analysis of Beams for Flexure and Shear - Beams Pre-Stressed With Straight, Concentric, Eccentric, Bent and Parabolic Tendons- Kern Line - Cable Profile - Design of PSC Beams (Rectangular and I Sections) Using IS 1343. Analysis and Design of Rectangular and I Beams for Shear. Introduction to Transmission Length and End Block (No Design and Analytical Problems).

UNIT – IV

DEFLECTIONS

Control of Deflections- Factors Influencing Deflections - Short Term Deflections ofUncracked Beams- Prediction of Long Time Deflections

UNIT – V		
Composite beams Different Types- Propped and Un-Propped- Stress Distribution- Differential Shrinkage- Analysis of Composite Beams.		
TEXT BOOKS:		
1. Prestressed Concrete by N. Krishna Raju, Tata McGraw Hill Publications 6th edition 2018 2. Prestressed concrete by N.Rajagopalan, Narosa Publishing House 2nd edition 2017		
REFRENCE BOOKS:		
1. Design of Prestressed Concrete Structures by T.Y. Lin & Ned H. Burns, John Wiley & Sons 3rd edition 2010 2. Prestressed Concrete Design by Praveen Nagrajan, Pearson publications, 2013. 3. Prestressed Concrete by Ramamuratam, Dhanpatrai Publications 2020 edition 4. BIS code on "prestressed concrete", IS: 1343 to be permitted into the examination Hall		
Online Learning Resources:		
https://archive.nptel.ac.in/courses/105/106/105106118/ **https://nptel.ac.in/courses/105106117**		